ÖZYEĞİN UNIVERSITY

EE473 - ENERGY SYSTEMS

Standalone solar panel system for the residential house in Kocaeli

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1 Introduction

This report explains the design and cost-effectiveness study of a standalone photovoltaic (PV) system. The system is designed to cover the energy needs of a normal family house in Kocaeli, Turkey. The design uses simulations made with PVsyst software [1].
The main goal of this project is to create an off-grid PV system that can reliably supply the estimated yearly energy needs of the house. The design work includes finding the right system size, choosing important parts based on project rules, simulating the system's performance with weather data from Meteonorm [2], and then doing a cost study to see if the investment is a good idea.
The suggested system uses PV modules from CW Enerji, a compatible MPPT charge controller from Victron Energy, and a Lithium Iron Phosphate (LFP) battery bank from BYD for energy storage. This setups goal is to ensure that energy is available when there is no sunlight and during times of low solar radiation. Cable sizing and related electrical loss checks are also important parts of this study.

1.1 Project Location

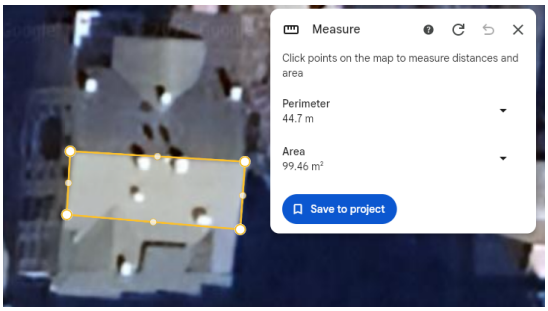
The chosen project site is a family house in Kocaeli, Turkey. The location and site details are:

- Latitude: 40.71° N
- Longitude: 29.93° E
- Altitude: 50 m
- Time Zone: UTC+3

A satellite image showing the exact location is in Figure 1. The general project location is shown in Figure 1a, and the possible area for putting the PV array on the roof, estimated at about 50 m^2, which is the half of the maximum area, is shown in Figure 1b.



(a) General project location in Kocaeli.



(b) Estimated roof area for PV installation.

Figure 1: Satellite imagery of the project site.

2 Methodology

The design work followed these steps:

1. **Load Assessment:** The average monthly and yearly energy use of the house was found from the monthly electricity bill of the house which is around 250 kWh.

2. **Component Selection:** PV panels, a charge controller, and a battery system were chosen based on project rules, like using Tier 1 or local manufacturers, specific panel watt, and suitable technology types.

3. **Simulation:** The PVsyst V8.0.9 software [1] was used to model the system's performance. Meteonorm 8.2 [2] weather data for Kocaeli (2001-2020) was used to create synthetic data. Important settings, like PV array direction, tilt, and detailed component information, were set in the simulation.

4. **System Sizing:** The needed number of PV panels and battery capacity were found by trying different options to meet the load, while balancing costs (Performance Ratio, Unused Energy) with performance (Solar Fraction).

5. **Cable Sizing and Loss Analysis:** Correct DC cable sizes were calculated based on current levels and acceptable power loss limits, aiming for less than 3% total DC losses.

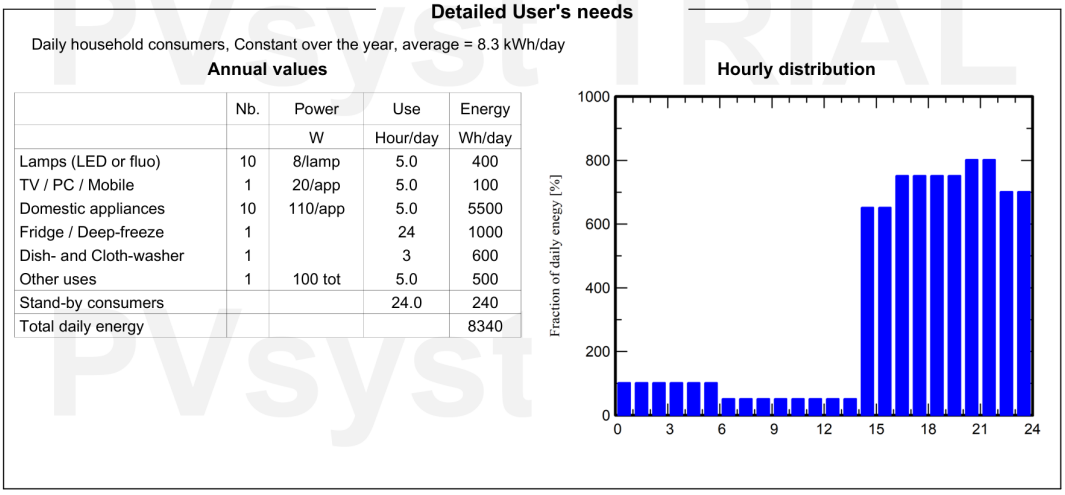
6. **Feasibility Analysis:** The total system cost was estimated, and the project's financial sense was checked, mainly by comparing it to the costs of grid connection to get power in an off-grid place.

3 System Design

The final suggested system is a standalone PV system with battery storage, which can operate for 3 days of autonomy and was designed from the simulation in PVsyst.

3.1 Load Profile (User's Needs)

The system is designed for an average monthly energy use of around 250 kWh, which means a yearly energy need of 3044.1 kWh (from PVsyst simulation). The hourly use pattern, based on normal home electricity use, is described in the figure 2.



	Nb.	Power	Use	Energy
		W	Hour/day	Wh/day
Lamps (LED or fluo)	10	8/lamp	5.0	400
TV / PC / Mobile	1	20/app	5.0	100
Domestic appliances	10	110/app	5.0	5500
Fridge / Deep-freeze	1		24	1000
Dish- and Cloth-washer	1		3	600
Other uses	1	100 tot	5.0	500
Stand-by consumers			24.0	240
Total daily energy				8340

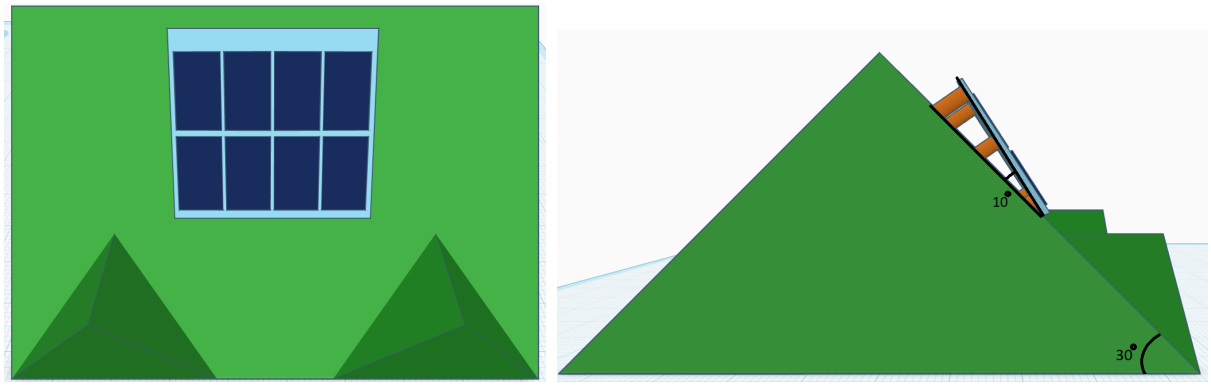
Figure 2: The hourly use pattern.

3.2 PV Array Configuration

The PV array details are:

- PV Panel Manufacturer: CW Enerji (Chosen as a local Turkish manufacturer)
- PV Panel Model: CWT580-144TN10
- Panel Nominal Power (STC): 580 W
- Number of Panels: 8
- Total Nominal Power (STC): $580 \times 8 = 4640$ W (4.64 kW)
- Array Configuration: 2 strings in parallel, each with 4 modules in series.
- Total Module Area: 20.7 m^2
- Orientation: Fixed Plane
- Tilt Angle: 40° (from horizontal)
- Azimuth Angle: 0° (due South)

These specific array details, such as the number of panels and their settings, were chosen after testing different options in the PVsyst software. The aim was to find the best setup for system performance and to reliably meet the energy needs. CW Enerji is a major Turkish PV panel maker [3]. The azimuth angle of 0° (due South) was selected because the project is in the Northern Hemisphere (Turkey), ensuring the panels face the sun's path for maximum daily energy capture. The tilt angle of 40° is close to the site's latitude (40.71°N), which is generally optimal for year-round energy collection, and particularly beneficial for improving solar energy gain during winter months when the sun is lower in the sky. This is important for a standalone system's reliability.
The solar panels are not placed flat on the existing roof. The roof already has a slope of 30°. However, for the solar panels to work best and capture the most sunlight all year, they need a final slope of 40°. This angle is more optimal because it's close to the area's latitude. To get this 40° slope, we use a custom mounting frame. This frame adds an extra 10° tilt to the panels on top of the roof's own slope. This setup helps the panels catch more sunlight, which is useful in winter when the sun is much less. You can see how this looks in Figure 3. Figure 3a shows a top view of the roof with the panels on the frame, and Figure 3b shows a side view where you can see the added tilt.



(a) Top view of the roof with solar panels placed on the mounting structure.

(b) Side view of the roof with solar panels placed on the mounting structure.

Figure 3: Image of the roof after installation.

3.3 Battery Storage System

The battery storage system details are:

- Battery Manufacturer: BYD (Chosen as a Tier 1 manufacturer)
- Battery Model: B-Box PRO 10
- Technology: Lithium Iron Phosphate (LFP)
- Number of Units: 4 (set in parallel to increase capacity)
- Nominal System Voltage: 51 V
- Total Nominal Capacity: 832 (at C10 rate)
- Total Stored Energy (from PVsyst): 38.3 kWh
- Minimum State of Charge (SOC): 10 % (Maximum Depth of Discharge of 90 %)

BYD is known battery maker [**BYDBattery**]. This large battery capacity, which holds 38.3 kWh of energy, is important to the system's autonomy. It shows us that the house can have electricity for few more days even if there is no sunlight, which is important for a reliable off-grid solution.

3.4 Charge Controller

The charge controller details are:

- Controller Manufacturer: Victron Energy (Chosen as a well-known manufacturer) [4]
- Controller Model: SmartSolar MPPT 250/70 48V
- Quantity: 1
- Technology: Maximum Power Point Tracking (MPPT)

This controller is the right size for the PV array voltage and current, and for the 48V battery system.

3.5 Cable Sizing and DC Losses

DC cables were chosen to reduce power loss and ensure safety. The PVsyst simulation shows an array current (I_{mpp}) of 27 A in use (total for 2 strings).

- **String Module Connections:** Each string has 4 modules in series, with a length of about 10 m to 15 m approximately so we chose 13 m. The string current (I_{mp}) is 13.5 A. A 2.5 mm^2 copper cable is chosen, which is fine for this current and helps reduce voltage drop.
- **Main Box to Battery:** The distance is about 3 m to 5 m approximately so we chose 4 m. The total array current from the junction box to the controller is around 27 A. A 10 mm^2 copper cable is chosen for this. This size is good for the current and reduces losses.
- **Simulated Ohmic Wiring Loss (Array):** 0.9 % at STC.
- **Simulated Module Mismatch Loss:** 2.0 % at MPP.
- **Total Estimated DC Losses (Ohmic + Mismatch):** About 0.9% + 2.0% = 2.9 %. This is below the project's maximum allowed DC loss of 3.0 %.

4 Simulation Results

The system's performance was simulated for one year using PVsyst V8.0.9 [1] with Meteonorm 8.2 [2] data for Kocaeli. The main yearly results from the simulation are in Table 1.

Table 1: Annual Simulation Results Summary.

Parameter	Value	Unit
Available Energy from Array	6539.0	kWh
Energy Supplied to User	3043.3	kWh
Energy Need of User	3044.1	kWh
Missing Energy	0.8	kWh
Loss of Load	0.1	%
Solar Fraction	99.97	%
Unused Energy	3313.2	kWh
Performance Ratio	40.76	%
Specific Production	656	kW h

The system has a very high Solar Fraction of 99.97 % and the system can store energy to supply for three consecutive days in case of no-sun phases. This means it reliably meets almost all of the yearly energy needs. The small amount of Missing Energy (0.8 kWh per year) means power is lost for only 0.1 % of the time. The monthly details in the PVsyst report, as shown in Table 2 and figure 4, indicate that this small unmet load happens mostly in December.

Table 2: Monthly and Annual Energy Balances

Month	GlobHor (kW h/m^2)	GlobEff (kW h/m^2)	E_Avail (kWh)	EUnused (kWh)	E_Miss (kWh)	E_User (kWh)	E_Load (kWh)	SolFrac (ratio)
Jan	53.8	90.4	391.0	128.9	0.0	258.5	258.5	1.0
Feb	62.6	80.3	347.6	83.2	0.0	233.5	233.5	1.0
Mar	102.1	118.0	507.6	234.6	0.0	258.5	258.5	1.0
Apr	141.1	145.2	613.6	347.7	0.0	250.2	250.2	1.0
May	188.1	173.2	713.9	439.9	0.0	258.5	258.5	1.0
June	203.4	177.3	722.9	456.8	0.0	250.2	250.2	1.0
July	203.1	181.4	728.9	456.1	0.0	258.5	258.5	1.0
Aug	182.7	183.4	739.6	465.3	0.0	258.5	258.5	1.0
Sep	133.0	151.7	620.9	356.3	0.0	250.2	250.2	1.0
Oct	85.0	111.6	463.9	190.9	0.0	258.5	258.5	1.0
Nov	57.1	88.2	372.2	129.1	0.0	250.2	250.2	1.0
Dec	45.3	73.2	316.9	24.3	0.8	257.8	258.5	1.0
Year	1457.1	1573.7	6539.0	3313.2	0.8	3043.3	3044.1	1.0

System Production

Useful energy from solar	3043.3 kWh/year	Perf. Ratio PR	40.76 %
Available solar energy	6539.0 kWh/year	Solar Fraction SF	99.97 %
Excess (unused)	3313.2 kWh/year		
Loss of Load		**Battery aging (State of Wear)**	
Time Fraction	0.1 %	Cycles SOW	98.6 %
Missing Energy	0.8 kWh/year	Static SOW	90.0 %

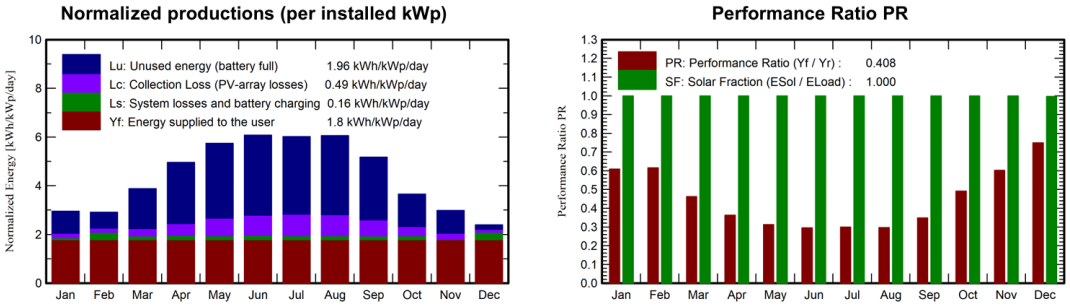


Figure 4: Overall losses and performance graphs.

The Performance Ratio (PR) of 40.76 % is normal for standalone systems made for high independence. The PR is lower than for typical grid-tied systems because of a lot of Unused Energy (3313.2 kWh, which is about 50.7 % of the array's available energy after system losses). This throwing away of energy is a normal part of off-grid system design. It happens when generation is more than the load and battery storage capacity in summer period, especially when the array is big enough for low-sun times. Because the system is designed to provide enough energy to use during winter period, this design leads system to create more than enough energy in summer period.

Key loss parts detailed in the PVsyst report are:

- **Sunlight Reflection Losses (IAM Factor):** Before sunlight enters the panels, about 2.2 % is lost because it reflects off the panel glass. This happens because the sun isn't always at a perfect angle to the panels. This is called the IAM (Incidence Angle Modifier) factor.
- **PV Module Performance Factors:** Even after light enters the panels, they don't convert all of it to electricity perfectly due to several reasons:
  - *Irradiance Level Effects:* The brightness of the sun affects how well panels work. If the sunlight is not at the ideal brightness right top of the panels, the panels produce slightly less electricity.
  - *Module Quality Loss:* A small amount of energy (−0.8 %) is lost due to very small imperfections in how the solar panels are made.
  - *Mismatch Losses:* Individual solar panels in the system are not all exactly identical. These small differences mean that when they work together, a little bit of energy (2.0 %) is lost.
- **Wiring Losses (Array Ohmic Losses):** When electricity flows through the DC cables from the panels, a small amount of energy (1.0 %) is lost as heat. This happens because the wires themselves have some resistance to electricity flow.
- **Battery System Losses:** Storing electricity in batteries and then taking it out again isn't perfectly efficient. About 9.7 % of energy is lost in this charging and discharging

process. Also, a very tiny amount (0.3 %) is lost just by keeping the energy stored in the batteries over time.

- **Unused (Extra) Energy:** Sometimes, especially on very sunny days, the solar panels produce more electricity than the house is using and more than the batteries can store (because they might already be full). This extra energy (49.4 % of the potential energy after some other losses) cannot be used by the system, so it's effectively lost. This is the biggest single reason why not all the sun's energy is used.
- **Temperature Losses:** Solar panels get hot when they sit in the sun. This heat makes them less efficient, so they produce less electricity than they would if they were cooler.

Overall, the main DC electrical losses from the array components (wiring losses at 0.9 % plus module mismatch losses at 2.0 %) total 2.9 %. This meets the project design target of keeping total DC losses below 3.0 %. The large amount of unused energy is a normal trade-off for making sure an off-grid system has enough power, especially during months with less sunshine.

5 Feasibility Study

This section looks at the estimated costs and overall analysis of installing the designed standalone PV system.

5.1 System Costs

A rough cost list for the system parts and installation is in Table 3. These costs are estimates based on limited online research (May 2025); actual prices require specific pricing from suppliers. Taxes (e.g., KDV) may not be included.

Table 3: Estimated System Cost Breakdown (Approximate, Q2 2025, TL)

Component	Specification	Qty	Unit Cost (TL)	Total Cost (TL)
PV Modules	CW Enerji CWT580-144TN10 [5]	8	6000	48,000
Charge Controller	Victron SmartSolar MPPT 250/70 48V [6]	1	18,000	18,000
Battery System	BYD B-Box PRO 10 (10.24 kWh nominal each)	4	70,000	280,000
Mounting Structure	Rooftop, fixed tilt	1	6000	6000
Cabling	DC (6 mm^2 Cu), connectors	1	3000	3000
Junction Box / Combiner	DC Box	1	1000	1000
Other Electrical (BoS)	Fuses, Breakers, etc.	1	2000	2000
Installation Labor	Qualified Installers	1	30,000	30,000
Permits/Misc (Optional)		1	1000	1000
Total Estimated System Cost (Approx.)				**389,000 TL**

5.2 Payback Period and Feasibility

Usually, you only build a system like this if you can't get good power from the normal grid or when you do not want to get connected to local provider. Comparing the system's high price to regular electricity bills makes it clear that having independent power, or power in a remote area, costs a lot extra.

Comparison with Grid Electricity Cost: The user consumes approximately 250 kWh per month, costing 520 TL.

- **Cost per kWh:** 520 TL / 250 kWh = 2.08 TL/kWh.
- **Annual Grid Cost:** 520 TL/month * 12 months = 6240 TL/year.
- **PV System Estimated Cost:** 389 000 TL (from Table 3).

Simple Payback Calculation (PV System vs. Grid Cost): This calculation shows how long it would take for the savings from not paying the grid electricity bill to equal the initial cost of the PV system, ignoring maintenance, battery replacement, and financing costs.

- Simple Payback = Total PV System Cost / Annual Grid Cost Savings
- Simple Payback ≈ 389 000 TL / 6240 TL/year ≈ **62 years**.

This extremely long payback period highlights that this standalone system is not financially competitive with the existing grid electricity price provided.

Feasibility Conclusion: From a purely financial perspective, based on the comparison with the current grid electricity cost of 2.08 TL/kWh, this standalone PV system is not feasible. The estimated initial investment of 389 000 TL is very high compared to the annual grid electricity cost of 6240 TL. The simple payback period of approximately 62 years is far beyond the expected lifetime of the system components, especially the batteries.
, the justification for installing such a system would rely entirely on other factors, such as:

- The location being truly off-grid with no possibility of grid connection.
- The cost of extending the grid to the location being significantly higher than the PV system cost because of the battery prices.
- Extreme unreliability of the existing grid supply, making energy independence a primary goal regardless of cost.
- Specific environmental goals or mandates outweighing the financial cost.

In simple terms, if the only goal is to save money compared to the current grid electricity price, this standalone system is too expensive, mostly because of the battery prices. It only makes financial sense if there are other strong reasons for building it, like needing power in a remote place where the grid doesn't reach, or wanting reliable power even if the grid often fails. For this specific project location where grid connection seems available, a grid-connected PV system (without batteries) would likely be a much more cost-effective and reasonable choice, as the high cost here comes mainly from the large battery bank needed for off-grid operation with a few days autonomy.

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[6] Amazon.com. "Victron energy smartsolar mppt tr solar charge controller (bluetooth) - 250v, 70 amp." Accessed May 2025. US Price ($478.55) used as indicator for Turkish market estimate (approx. 18000 TRY estimate derived). (2025), [Online]. Available: https://www.amazon.com/Victron-SmartSolar-MPPT-250-70-Tr/dp/B08HS516N9.